

- Fetching spatial data in internal format:

Fetching geometry values using internal format can be useful in table-to-table transfers:

```
CREATE TABLE geom2 (g GEOMETRY) SELECT g FROM geom;
```

- Fetching spatial data in WKT format:

The `ST_AsText()` function converts a geometry from internal format to a WKT string.

```
SELECT ST_AsText(g) FROM geom;
```

- Fetching spatial data in WKB format:

The `ST_AsBinary()` function converts a geometry from internal format to a `BLOB` containing the WKB value.

```
SELECT ST_AsBinary(g) FROM geom;
```

11.4.9 Optimizing Spatial Analysis

For `MyISAM` and `InnoDB` tables, search operations in columns containing spatial data can be optimized using `SPATIAL` indexes. The most typical operations are:

- Point queries that search for all objects that contain a given point
- Region queries that search for all objects that overlap a given region

MySQL uses **R-Trees with quadratic splitting** for `SPATIAL` indexes on spatial columns. A `SPATIAL` index is built using the minimum bounding rectangle (MBR) of a geometry. For most geometries, the MBR is a minimum rectangle that surrounds the geometries. For a horizontal or a vertical linestring, the MBR is a rectangle degenerated into the linestring. For a point, the MBR is a rectangle degenerated into the point.

It is also possible to create normal indexes on spatial columns. In a non-`SPATIAL` index, you must declare a prefix for any spatial column except for `POINT` columns.

`MyISAM` and `InnoDB` support both `SPATIAL` and non-`SPATIAL` indexes. Other storage engines support non-`SPATIAL` indexes, as described in [Section 13.1.15, “CREATE INDEX Statement”](#).

11.4.10 Creating Spatial Indexes

For `InnoDB` and `MyISAM` tables, MySQL can create spatial indexes using syntax similar to that for creating regular indexes, but using the `SPATIAL` keyword. Columns in spatial indexes must be declared `NOT NULL`. The following examples demonstrate how to create spatial indexes:

- With `CREATE TABLE`:

```
CREATE TABLE geom (g GEOMETRY NOT NULL SRID 4326, SPATIAL INDEX(g));
```

- With `ALTER TABLE`:

```
CREATE TABLE geom (g GEOMETRY NOT NULL SRID 4326);
ALTER TABLE geom ADD SPATIAL INDEX(g);
```

- With `CREATE INDEX`:

```
CREATE TABLE geom (g GEOMETRY NOT NULL SRID 4326);
CREATE SPATIAL INDEX g ON geom (g);
```